

Ananya

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SUMMARY

ML engineer with two years at NVIDIA and three applied ML internships spanning empirical work on computer vision, generative models, and language models. Hands-on experience training, debugging, and evaluating deep learning models against real, noisy data rather than clean benchmarks, and iterating quickly based on evaluation signals rather than treating models as fixed artifacts. Comfortable owning work end-to-end: data, training, evaluation, and re-iteration.

EDUCATION

University of Edinburgh

MSc in Artificial Intelligence

Edinburgh, UK

Sept 2026 – Sept 2027

Indian Institute of Technology (IIT) Roorkee

B.Tech. in Electrical Engineering

India

2020 – 2024

- **Thesis:** “Deep Learning Models for Time-Series Fault Detection in Power Grids” (A) — signal-to-image conversion with CNN/Transformer modeling for detection of anomalous patterns in sensor data.

EXPERIENCE

NVIDIA — *Software Engineer, ML Platform Infrastructure*

India

Jul 2024 – Sept 2026

- Ran empirical experiments and benchmarks for a research team on a follow-up to NVIDIA’s **Gated DeltaNet-2** language-model architecture, evaluating variants against baselines and debugging training runs on large GPU clusters, direct experience training, evaluating, and iterating on models based on clear performance signals.
- Owned firmware recovery pipelines (OpenBMC/PLDM) across **HGX**, **GB200**, **Vera Rubin**, and **Bali/Moyo**, production infrastructure with zero tolerance for silent failure.
- Redesigned the Vera Rubin CPU recovery flow from I2C to USB RCM as sole owner, cutting node recovery time by **700×** and reducing downtime for production DL training runs.

JM Financial — *Data Science Intern, Applied ML*

India

Dec 2022 – Mar 2023

- Trained and compared several models end to end on a noisy, real-world Google Keywords dataset, iterating on approaches for accuracy and presenting findings to stakeholders.

Samsung R&D — *Software Engineering Intern, Generative ML*

India

May 2023 – Jul 2023

- Built and evaluated GAN-based image models for Samsung’s FRAME TV; scoped technical feasibility with PM and engineering and ran milestone demos to track delivery against plan.

IISc Bangalore, Video Analytics Lab — *Research Intern*

India

Jun 2022 – Sep 2022

- Modified, combined, and empirically compared several state-of-the-art approaches to StyleGAN latent-space disentanglement; worked independently and presented findings to the lab.

SELECTED PROJECTS & ACHIEVEMENTS

- **Robotic EV-charging detection** (*2nd Runner-Up, Inter-IIT Tech Meet — Jaguar Land Rover Track, 2023*): Built the computer-vision **detection and localization** component that let a robotic charging arm locate and align with a car’s charging port autonomously, an end-to-end perception system trained and evaluated against real-world constraints.
- **SiamCAR/SiamMOT vehicle tracking** adapted for Indian road footage, hands-on experience with real-time object tracking on noisy, real-world driving data.
- **Pix2Pix-based conditional GAN** for image colorization, applied generative modeling experience.
- **Transformer / language model from scratch** (*2023–2024*): Built an autograd engine and a character-level language model (bigram → MLP → transformer) from first principles to understand attention and backpropagation at a low level.
- **Water quality prediction** (*Winner, Intel OneAPI / NSM AI Hackathon, 2023*): Built a stacked-ensemble model; placed 1st nationwide.

SKILLS

ML/DL: PyTorch (primary), TensorFlow, scikit-learn, computer vision (detection, tracking, segmentation), GANs, CNNs, transformers/LLMs, LSTM/seq2seq, empirical model training & evaluation

Systems: Python (production), C++, SQL, pandas/NumPy, Git, distributed GPU systems, data pipelines, debugging large-scale training runs